

Vision-Language-Action Models for Semantic Plastic Waste Sorting with Transparent Object Perception

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Abstract

Transparent plastics defeat the active infrared depth sensors on which robotic sorting depends, and no corpus of robot demonstrations exists for transparent waste—jointly blocking the application of modern Vision-Language-Action (VLA) models to a commercially significant recycling problem. This dissertation investigates whether a 7-billion-parameter VLA (OpenVLA) can be adapted to transparent-waste grasp targeting using exclusively synthetic supervision, alongside a learned software repair of the depth-sensing failure.

A three-stage pipeline was designed and evaluated. ReSort-IT, a procedural generator, composites transparent-object renders onto varied backgrounds and derives per-image grasp targets and discretised action tokens, producing 44,077 training and evaluation triplets in its final mixed iteration, with background-identity-controlled data splits. A dual-head DeepLabV3 network predicts surface normals and segmentation from RGB, feeding a perspective-corrected Poisson solver that reconstructs metric depth over corrupted regions. The VLA is adapted with Low-Rank Adaptation (0.22% trainable parameters) under 4-bit quantisation on a single GPU.

Results are reported with statistical testing throughout. The perception module attains 19.6° mean normal error and 91% IoU in distribution, generalising to unseen shapes in proportion to geometric similarity. A four-way ablation shows the depth solver matches a strong inpainting baseline given perfect normals, and isolates a 0.28 m RMSE cost attributable solely to perception error. An initial training configuration collapsed into shortcut memorisation concealed by 97.7% token accuracy and zero training loss; with per-sample target diversity, background decorrelation, and seam removal enforced, the policy achieved genuinely image-conditioned predictions whose predictions vary with image content, unlike the constant-output memorising baseline (mean error 27.6 bins, ≈ 11 native/composite training mix, and encoder-native resolution—nearly tripled the exact on-target grasp rate from 2.6% to 7.2%, with a diagnostic type breakdown (11.3% on native frames, 3.1% on composites) directly attributing the improvement to the restoration of transparency-specific visual cues that compositing removes. End-to-end throughput is 0.26 FPS, bounded by the CPU depth solver (65%) rather than the language model.

The work contributes a documented diagnosis of action-supervision shortcut learning, a quantified perception-to-depth error propagation measurement, and a costed feasibility map for synthetic-only VLA training on transparent waste.

Keywords: Vision-Language-Action models; transparent object perception; robotic waste sorting; synthetic data generation; depth reconstruction; parameter-efficient fine-tuning.

Chapter 1

Introduction

1.1 Motivation

Automated waste sorting is a prerequisite for economically viable recycling at scale. Robotic sorting systems have matured considerably for opaque materials, where colour, texture, and reliable depth sensing support both recognition and manipulation [1, 2]. Transparent plastics, however, remain a persistent blind spot. Polyethylene terephthalate (PET) bottles and similar clear containers constitute a large and valuable fraction of the recyclable waste stream, yet they defeat the active infrared depth sensors on which most robotic manipulation pipelines depend: structured light refracts through and reflects off transparent surfaces, so the sensor returns missing or corrupted depth measurements precisely over the objects that must be grasped [3].

At the same time, a new class of manipulation policy has emerged. Vision-Language-Action (VLA) models such as RT-2 [4] and OpenVLA [5] couple large pre-trained vision-language backbones to robot action decoding, promising semantic flexibility that conventional sorting pipelines lack: the ability to follow natural-language instructions and to draw on web-scale visual knowledge when confronting object categories absent from their robot training data. These properties are directly relevant to waste sorting, where the variety of incoming items is effectively unbounded. However, VLAs are trained on large corpora of real robot demonstrations, and no such corpus exists for transparent waste manipulation; collecting one at the required scale is impractical within realistic project constraints.

This dissertation sits at the intersection of these two problems. It asks whether a modern VLA can be adapted to transparent-object sorting *without* real robot demonstrations, using procedurally generated synthetic data, and whether the depth-sensing failure can be repaired in software by a learned perception front-end, so that the combined system produces metrically meaningful grasp targets.

1.2 Aim and Objectives

The aim of this work is to design, implement, and quantitatively evaluate a cascaded software pipeline for language-conditioned grasp-target prediction on transparent plastic waste, in which all training supervision is generated synthetically.

This aim is decomposed into four objectives:

1. To construct an automated data generation framework (ReSort-IT) that composites transparent-object assets onto varied backgrounds and derives per-image grasp targets and action-token labels without manual annotation.
2. To train and evaluate an RGB-only perception module that predicts surface normals and segmentation masks for transparent objects, and to quantify its generalisation to object shapes never seen in training.
3. To reconstruct metric depth over the corrupted transparent regions from the predicted geometry, and to establish through controlled ablation whether this reconstruction outperforms geometry-unaware baselines.
4. To fine-tune a 7-billion-parameter VLA (OpenVLA) on the synthetic data using parameter-efficient adaptation, and to measure the spatial accuracy of its grasp-target predictions, the degree to which that accuracy supports physically plausible grasps, and the computational cost of the full pipeline.

The corresponding research questions are:

- RQ1** Can procedurally composited synthetic data provide sufficient supervision for a VLA to learn image-conditioned spatial grounding of transparent objects, and what properties of the data determine success or failure?
- RQ2** How accurately does an RGB-only convolutional network recover the surface geometry of transparent objects, and how does that accuracy degrade on unseen object categories?
- RQ3** Does surface-normal-guided depth reconstruction improve on geometry-unaware inpainting, and how does error in the predicted normals propagate into the reconstructed depth?
- RQ4** Is the integrated pipeline computationally feasible for deployment on a moving conveyor, and if not, which component bounds its throughput?

[PLACEHOLDER — RQ5 to be added once the semantic-reasoning experiment is complete: Does the fine-tuned policy condition its spatial predictions on the language instruction, selecting between different object categories present in the same scene?]

1.3 Contributions

The principal contributions of this dissertation are: (i) ReSort-IT, a procedural pipeline that converts static transparent-object renders into VLA-ready image-language-action triplets, together with an empirical analysis of the data properties (target diversity, background decorrelation, seam removal) that separate genuine spatial learning from memorisation; (ii) a quantified error-propagation analysis linking surface-normal prediction error

to downstream metric depth error, obtained through a four-way controlled ablation with statistical significance testing; and (iii) an honest end-to-end characterisation of a synthetic-only VLA sorting pipeline, including per-stage latency and a grasp-feasibility proxy, that identifies the current bottlenecks and their implications for real deployment.

1.4 Dissertation Structure

Chapter 2 reviews the literature on robotic waste sorting, transparent-object perception, and Vision-Language-Action models, and locates the research gap this work addresses. Chapter 3 details the methodology: the ReSort-IT data pipeline, the dual-head perception network and Poisson depth solver, the LoRA fine-tuning protocol, and the evaluation design. Chapter 4 presents the quantitative results for each component and for the integrated system. Chapter 5 discusses the findings in relation to the research questions and the wider literature. Chapter 6 concludes, states the limitations of the study, and sets out recommendations for future work.

Chapter 2

Literature Review

2.1 Introduction

Plastic waste sorting faces two coupled challenges: a global recycling shortfall and a long-standing robotic automation bottleneck. Globally, less than 10% of plastic waste is appropriately recycled, with a large proportion lost to inadequate sorting infrastructure [6]. Industrial Material Recovery Facilities (MRFs) still depend heavily on near-infrared (NIR) optical sensors and rigid rule-based controllers that struggle with material variability, deformation, and contamination [2, 7]. Vision-Language-Action (VLA) models offer a different premise: by integrating visual perception, language instruction, and action generation in a single architecture pre-trained on internet-scale data, they promise generalisation across the diverse morphologies of real waste without hard-coded rules per plastic type [8, 9].

Translating that promise to sorting lines, however, collides with several unsolved problems, two of which this dissertation isolates: transparent plastics defeat the depth sensors on which manipulation pipelines rely, and no corpus of robot demonstrations exists on which to fine-tune a VLA for this domain. This chapter reviews the classical sorting literature (Section 2.2), scene understanding for robotic sorting with emphasis on transparent-object perception (Section 2.3), the VLA paradigm and its documented spatial limitations (Sections 2.4–2.5), and synthetic-data strategies for training such systems (Section 2.6). Section 2.7 consolidates the technical gaps, and Section 2.8 states the research aim and questions that follow from them.

2.2 Classical Plastic Sorting: Sensing and Classification

Lubongo et al. [2] survey current MRF sorting technology, documenting three broad approaches: mechanical separation by density, electromagnetic extraction of ferrous materials, and optical sorting using spectral sensors. Among these, NIR spectroscopy has become the industrial standard: it permits contact-free identification of polymer types from their molecular absorption spectra, and commercial NIR systems achieve throughputs of 60–80 picks per minute with classification accuracies above 99% under controlled conditions [2].

Classical systems nonetheless face structural limitations. Standard NIR equipment cannot identify black plastics: the carbon-black pigment absorbs nearly all incident light

across the 0.9–1.7 μm range, rendering these items invisible to the sorter [7]. Mid-wave infrared hyperspectral imaging (3–5 μm), where plastics exhibit distinct absorption features regardless of colour, has emerged as the most effective response [7], at considerably higher hardware and processing cost. Contamination is a second failure mode: food residue and labels degrade the accuracy of both NIR and visual sensors [6, 7], and rule-based systems lack the contextual reasoning to recognise a contaminated PET bottle as PET.

On the machine-vision side, RGB-based classification has progressed steadily. Bobulski and Kubanek [10] benchmark CNNs for polymer classification from low-cost RGB cameras, reporting 74% cross-validated accuracy for a lightweight 15-layer network and 86% for MobileNet-v2—sufficient for broad material categories but not fine-grained polymer discrimination. One-stage detectors such as YOLO variants have become the dominant choice for real-time conveyor detection [11]. Kroell et al. [7] conclude from their systematic review that while deep learning has improved classification flexibility, distinguishing chemically distinct but visually identical polymers—clear PET from clear PVC in particular—remains unsolved by RGB methods alone, making spectral data a necessity wherever chemical accuracy is required [2, 12].

A further, less discussed limitation is geometric: spectroscopic and RGB classification answer *what* is on the belt but not *where to grasp it*. Converting a detection into a successful pick on cluttered, deformed waste is repeatedly identified as harder than the classification itself [2]. Koskinopoulou et al. [1] present an integrated vision-based robotic sorting system on an industrial line—combining deep-learning categorisation, grasp selection, and robotic separation, with reported sorting accuracy above 90%—which exemplifies the state of practice and, equally, its constraints: the perception stack presumes usable depth sensing, and extending the system to new categories or new sorting instructions requires retraining and re-engineering rather than issuing a new command.

2.3 Semantic Scene Understanding for Robotic Sorting

Transforming raw sensor data into actionable representations for sorting rests on three components: open-vocabulary detection, depth reconstruction, and affordance prediction.

Traditional detectors require a fixed category vocabulary defined at training time, making them brittle against novel packaging. VoxPoser [13] demonstrates a more flexible alternative by chaining an open-vocabulary detector (OWL-ViT) with the Segment Anything Model, identifying and masking arbitrary objects described in natural language—“the crushed translucent bottle”—without labelled examples of that object. This capability motivates language-conditioned sorting, but VoxPoser’s downstream machinery composes LLM-generated 3D value maps over voxel grids and therefore *presumes reliable 3D perception*—precisely what transparent objects deny.

Depth reconstruction for transparent objects is the second component. Active infrared sensors such as the Intel RealSense series project structured light that refracts through or transmits entirely across transparent PET and PVC, producing depth maps with severe holes and distortions exactly over the objects most common in plastic waste [3]. ClearGrasp [3] addresses this directly: from the RGB image it predicts surface normals,

transparent-object masks, and occlusion boundaries via three parallel DeepLabV3-based networks, then solves a global optimisation that reconstructs depth over the corrupted regions, anchored to the reliable measurements of the surrounding opaque scene. Trained purely on a 45,000-image physically based synthetic dataset (which also supplies the assets used in the present work), ClearGrasp raised parallel-jaw grasp success on real transparent objects from a 12% raw-depth baseline to 74%, establishing that the RGB-to-geometry route bridges the sim-to-real gap for this material class. Two properties of ClearGrasp are inherited here—the decomposition into RGB-based geometric prediction plus optimisation-based completion, and synthetic supervision—while one is deliberately departed from: ClearGrasp terminates in a task-specific grasp network with no semantic or language interface.

Affordance prediction closes the loop from perception to action. VoxPoser’s LLM-composed value maps assign reward to actionable regions such as a bottle neck [13]. RoboPoint [14] instead trains a vision-language model to predict 2D spatial coordinates for grasp targets directly, trading 3D resolution for inference efficiency—an approach structurally close to the coordinate-token prediction evaluated in this dissertation. SpatialVLA [15] extends the idea by injecting egocentric 3D representations into the VLA action-token space to better resolve spatial relations in clutter.

2.4 Vision-Language-Action Models

The VLA paradigm restructures robotic system design: rather than composing separate modules for perception, planning, and control, a single model learns an end-to-end mapping from visual observations and language instructions to actions [8, 9]. Surveys catalogue over eighty VLA models published since 2022, categorised by action-generation strategy—autoregressive token prediction, diffusion, and flow matching [16–18]—and by functional role: sensorimotor models, world models, and affordance-based models, the latter particularly suited to unstructured environments because the generated action is decoupled from a specific robot’s kinematics [9].

RT-2 [4] established the token approach at scale: continuous end-effector displacements and gripper commands are discretised into 256 bins and generated as text, and co-fine-tuning on web-scale vision-language data alongside robot trajectories prevents catastrophic forgetting of semantic knowledge while acquiring physical skill. RT-2’s headline capability—manipulating objects never seen in the robot data by drawing on web knowledge—is precisely the property that makes VLAs attractive for waste streams of unbounded variety. RT-2 is, however, closed and computationally heavy (its 55B variant runs at 1–3 Hz on multi-TPU cloud hardware; the 5B variant at ~ 5 Hz [4]).

OpenVLA [5] made the paradigm accessible: a 7B-parameter open-source model pairing a fused SigLIP–DINOv2 vision encoder with a Llama-2 backbone, trained on 970k real-robot episodes, adopting the same 256-bin action tokenisation. Critically for resource-constrained settings, Kim et al. [5] demonstrate effective adaptation with Low-Rank Adaptation [19] from as few as tens of demonstrations, and quantised operation [20] brings fine-tuning within reach of a single high-memory GPU—the configuration adopted in Chap-

ter 3. The choice of tokenisation itself is an active research area [18]; uniform binning is the simplest scheme, and its 256-level granularity bounds achievable spatial precision. Complementary directions add intermediate reasoning (visual chain-of-thought in CoT-VLA [21]), explicit 3D world models [22], or additional physical sensing modalities (OmniVLA [23]).

2.5 Spatial Grounding: a Documented Weakness

A consistent finding across recent evaluations is that VLAs, and the vision-language models beneath them, are semantically strong but metrically weak. SpatialBench [24] documents systematic deficits in localisation and spatial-relation reasoning even where recognition is nearly perfect. The affordance and representation work of Section 2.3—RoboPoint’s dedicated pointing supervision [14], SpatialVLA’s 3D encodings [15], 3D-VLA’s generative world model [22]—can be read collectively as responses to this weakness: spatial grounding does not emerge from scale alone but must be engineered through data, representation, or architecture. For sorting, the implication is sharp: recognising a bottle is insufficient; the policy must emit coordinates accurate to a gripper’s tolerance. This dissertation contributes evidence from an unusual angle—measuring how far purely synthetic, composited 2D supervision pushes spatial grounding in a stock OpenVLA, and characterising what happens when the data is insufficient.

2.6 Synthetic Data and Its Hazards

Collecting real robot demonstrations is expensive; for transparent waste no public corpus exists. Two synthetic strategies dominate. Physically based rendering, as in ClearGrasp [3], yields high-fidelity images with exact ground truth at the cost of 3D assets and rendering infrastructure. Compositing—pasting segmented foregrounds onto varied backgrounds, as in the ReSort-IT augmentation methodology of Koskinopoulou et al. [1] adopted and extended in this work—is far cheaper, at the cost of physical implausibility: composites lack correct shadows and contact regions and, for transparent objects, refraction of the true background through the object. Domain randomisation of pose, scale, photometry, and backgrounds is the standard mitigation, forcing reliance on object-intrinsic cues.

The literature is comparatively quiet on a hazard specific to composited *action* supervision: if target labels correlate with background identity, or compositing artefacts such as sharp alpha seams are visible, a high-capacity model can minimise training loss through shortcuts carrying no spatial understanding—and, because most dimensions of an action string are near-constant, aggregate token metrics can conceal the failure entirely. Chapter 4 documents a clean instance of this failure and its diagnosis, contributing an empirical data point on what composited data must control for when the supervised quantity is a spatial action rather than a class label.

2.7 Technical Gaps for VLA-Based Plastic Sorting

Synthesising the review, four structural mismatches separate current VLA systems from deployment in plastic sorting.

Gap 1: Inference latency. Autoregressive VLAs generate actions token by token and cannot reach the control rates of reactive manipulation: RT-2 runs at 1–5 Hz depending on scale [4], OpenVLA at roughly 5 Hz on a high-end GPU [5], and Kawaharazuka et al. [9] identify this as the central deployment barrier. Sorting lines compound the problem with continuously moving targets.

Gap 2: The embodiment and domain gap. VLA training data consists overwhelmingly of clean, well-lit household manipulation on specific arms; industrial waste—deformed, contaminated, cluttered, and moving—is out of distribution both visually and kinematically [9]. Classical pipelines such as Koskinopoulou et al. [1] remain the benchmark on real waste precisely because they were engineered in-domain.

Gap 3: Data scarcity for transparent waste. No demonstration corpus exists for transparent-waste manipulation, and transparent objects additionally invalidate the depth channel that RGB-D policies and 3D affordance methods presume (Section 2.3). Synthetic generation is the only accessible route, and Section 2.6 identified its under-documented shortcut hazards for action supervision.

Gap 4: Absence of spectral integration. Current VLAs operate on RGB or RGB-D inputs and cannot chemically discriminate visually identical polymers such as clear PET and clear PVC [2]; OmniVLA’s multi-sensor representations [23] have not been applied to NIR polymer classification. This dissertation operates deliberately *within* that constraint: its transparent target categories are drawn from the ClearGrasp object set and are distinguished morphologically, chemical identification being excluded from scope and identified as future work.

Table 2.1 summarises the principal approaches against these gaps.

2.8 Research Gap, Aim, and Questions

No prior work, to the author’s knowledge, combines the elements this review has assembled: a VLA fine-tuned for transparent-waste grasp targeting using exclusively composited synthetic supervision, evaluated jointly with a learned depth-repair front-end, with the error propagation between the two quantified and the shortcut hazards of composited action data documented. Within the resource envelope of an MSc project—one GPU, no demonstration corpus, no guaranteed robot access—the deployable-system question of the original proposal is deliberately narrowed to the prior question on which it depends: *under what data conditions does synthetic-only VLA adaptation produce genuine spatial grounding on transparent objects, and how do the perception and policy errors compose?*

The aim of this work is therefore to design, implement, and quantitatively evaluate a cascaded pipeline for language-conditioned grasp-target prediction on transparent plastic waste, trained entirely on synthetic data, answering:

Table 2.1: Summary of key approaches in automated plastic sorting and robotic manipulation.

Study	Modality	Architecture	Key contribution and limitation
Lubongo et al. [2]	NIR spectroscopy	Rule-based	Industrial standard, $\sim 99\%$ accuracy under controlled conditions; blind to black plastics; no grasp geometry.
Kroell et al. [7]	RGB vision	CNN / YOLO	Flexible visual classification; cannot separate chemically distinct, visually identical polymers (PET vs. PVC).
Koskinopoulou et al. [1]	RGB(-D)	Deep classical pipeline	Integrated robotic sorting on an industrial line; presumes valid depth; retraining required per category.
Sajjan et al. [3]	RGB-D	DeepLabV3 + optimisation	Transparent depth reconstruction (grasp success $12\% \rightarrow 74\%$); no semantic or language interface.
Zitkovich et al. [4]	RGB + text	VLA (co-fine-tuned)	Web-knowledge transfer to control; closed; 1–5 Hz inference.
Kim et al. [5]	RGB + text	Prismatic VLM + Llama-2	Open-source VLA; LoRA-adaptable on one GPU; inherits latency and spatial-precision limits.
Guo et al. [23]	Multi-sensor	Sensor-fused VLA	Non-RGB modalities in VLA representations; unvalidated for NIR sorting.

RQ1 Can procedurally composited synthetic data provide sufficient supervision for a VLA to learn image-conditioned spatial grounding of transparent objects, and what properties of the data determine success or failure?

RQ2 How accurately does an RGB-only network recover the surface geometry of transparent objects, and how does that accuracy degrade on unseen object categories?

RQ3 Does surface-normal-guided depth reconstruction improve on geometry-unaware inpainting, and how does error in the predicted normals propagate into reconstructed depth?

RQ4 Is the integrated pipeline computationally feasible for conveyor deployment, and if not, which component bounds its throughput?

[PLACEHOLDER — add RQ5 here once the semantic-reasoning experiment is under way: Does the fine-tuned policy condition its spatial predictions on the language instruction, selecting between different object categories present in the same scene?]

The following chapter details the methodology through which these questions are answered.

Chapter 3

Methodology

This chapter describes the design and implementation of a software pipeline for the semantic sorting of transparent plastic waste. The system addresses two obstacles that jointly prevent the direct application of existing robotic manipulation methods to this domain: active infrared depth sensors return corrupted or missing measurements over transparent surfaces, and no large-scale dataset of language-annotated robot demonstrations exists for transparent waste items.

The pipeline is organised as a cascade of three components, developed as separate work packages. Work Package 1 (WP1) is a procedural data generation framework, referred to as ReSort-IT, which synthesises image–language–action training triplets from existing transparent-object assets without requiring physical robot demonstrations. Work Package 2 (WP2) is a multi-task convolutional network that predicts surface normals and segmentation masks from RGB input, feeding a depth reconstruction solver that repairs the corrupted sensor stream. Work Package 3 (WP3) fine-tunes a pre-trained 7-billion-parameter Vision-Language-Action model [5] on the synthetic data using parameter-efficient adaptation. The chapter closes with the evaluation methodology used to assess each component and the integrated system.

A cascaded design was chosen over end-to-end training for three reasons. First, it decouples metric depth recovery from semantic reasoning, so each component can be evaluated, debugged, and replaced independently. Second, it keeps the large language-model backbone isolated from geometric computation, for which it is poorly suited. Third, it permits the perception front-end and the policy to be trained on different data sources, which is necessary because ground-truth geometry (normals, depth) and action labels come from different generation processes.

3.1 Work Package 1: Synthetic Data Generation (ReSort-IT)

3.1.1 Source Assets

Foreground objects were sourced from the ClearGrasp synthetic dataset [3], which provides photorealistic renders of transparent plastic objects with per-pixel segmentation masks, surface normals, and depth. The `square-plastic-bottle` category, comprising 11,702

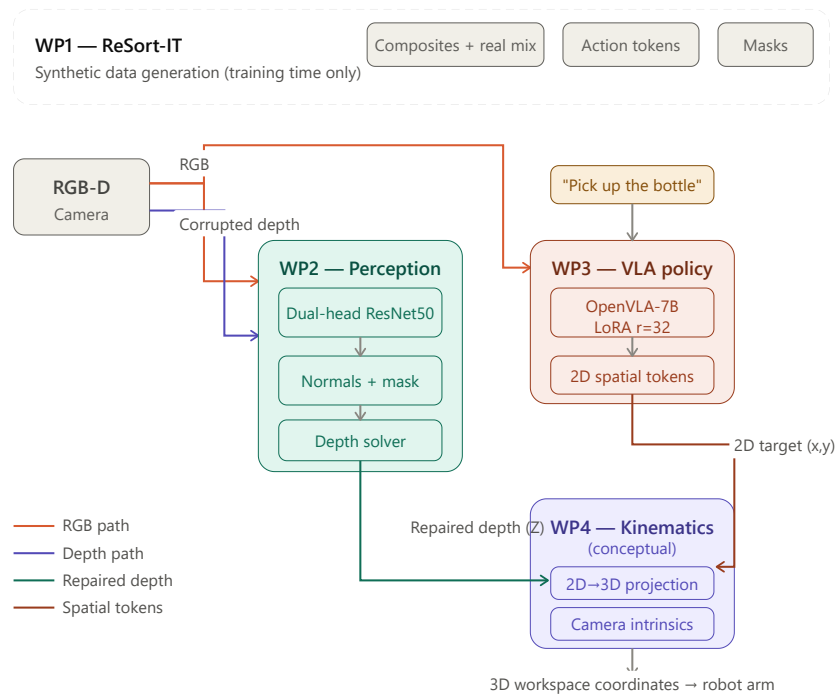


Figure 3.1: End-to-end system architecture. The WP2 perception module provides repaired metric depth, which combines with the WP3 VLA policy’s 2D spatial tokens to yield 3D workspace coordinates via WP4 kinematics.

RGB-mask pairs, was used to train the manipulation policy. Background plates were drawn from a separately collected pool of 153 texture images spanning industrial and domestic surfaces (conveyor rubber, wood, concrete, fabrics, and patterned materials).

A deliberate scoping decision was made to restrict policy training to a single object category. This isolates the central research question—whether a VLA can learn spatial grounding of transparent objects from composited synthetic data—from the separate question of multi-class generalisation. The remaining ClearGrasp categories were reserved as held-out data for evaluating the generalisation of the perception module (Section 3.4).

3.1.2 Grasp Target Derivation

Each ClearGrasp render may contain several disconnected transparent objects. Deriving a grasp target from the centroid of the full mask fails in this situation: averaging over multiple objects places the target in the empty space between them, a failure mode observed during early development and referred to here as the *empty-centre* problem. On a physical sorting line this would direct the end-effector into the bare conveyor surface.

To resolve this, individual objects are first isolated by connected-component analysis. The binary mask $\mathbf{M}$ is decomposed into external contours,

$$\mathcal{C} = \{c_1, \dots, c_n\} = \text{findContours}(\mathbf{M}), \quad (3.1)$$

and one contour c_i is selected uniformly at random as the grasp target for that composite; the remaining objects are retained in the frame as clutter. The target coordinate is then the centre of mass of the selected contour, computed from its image moments,

$$M_{pq} = \sum_{(x,y) \in c_i} x^p y^q, \quad c_x = \frac{M_{10}}{M_{00}}, \quad c_y = \frac{M_{01}}{M_{00}}, \quad (3.2)$$

which places the ground-truth target on solid object material rather than in the gap between objects for convex shapes. The centroid guarantee does not, however, hold for concave objects: for crescent-shaped or crushed items, the centre of mass can fall outside the object entirely. An audit of the first generated dataset (Section 4.4.1) found that 10% of targets suffered exactly this defect. The final generator therefore validates every centroid with a point-in-polygon test,

$$\text{pointPolygonTest}(c_i, (c_x, c_y)) \geq 0, \quad (3.3)$$

and, where the test fails, substitutes the contour’s most-interior point—the argmax of the L2 distance transform of the filled contour—which lies on object material by construction. A runtime assertion additionally verifies every emitted label against its saved object mask, so label validity in the final dataset is 100% by construction rather than by expectation.

3.1.3 Compositing and Domain Randomisation

Each source image is composited onto randomly selected backgrounds three times, with independently sampled transformations, yielding 35,103 composites in total for the initial baseline dataset (Run 1). The following randomisations are applied to every composite:

- **Scale.** The foreground is resized so that its width is a fraction $s \sim \mathcal{U}(0.25, 0.75)$ of the 640-pixel background width, covering both distant and close object appearances.
- **Geometric transforms.** A horizontal flip is applied with probability 0.5, followed by a rotation of $\theta \sim \mathcal{U}(-15^\circ, 15^\circ)$. The mask is re-thresholded after rotation to remove interpolation artefacts.
- **Photometric jitter.** Foreground contrast is scaled by $\alpha \sim \mathcal{U}(0.6, 1.4)$ and brightness shifted by $\beta \sim \mathcal{U}(-40, 40)$; in HSV space, hue is shifted by up to $\pm 10^\circ$ and saturation scaled by $\mathcal{U}(0.7, 1.3)$.
- **Edge softening.** The binary mask is blurred with a 7×7 Gaussian kernel before alpha blending. This removes the sharp compositing seam at the object boundary, which would otherwise provide the network with a localisation shortcut unrelated to object appearance.
- **Sensor noise.** Zero-mean Gaussian noise with $\sigma \sim \mathcal{U}(1, 5)$ intensity levels is added to the final composite.
- **Position.** The composite location is sampled uniformly subject to the object remaining fully in frame, and independently of the background identity, so that no correlation exists between background appearance and target position.

Alpha blending with the softened mask $\tilde{\mathbf{M}}$ produces the final composite:

$$\mathbf{I}_{\text{comp}} = \mathbf{I}_{\text{fg}} \odot \frac{\tilde{\mathbf{M}}}{255} + \mathbf{I}_{\text{bg}} \odot \left(1 - \frac{\tilde{\mathbf{M}}}{255}\right). \quad (3.4)$$

3.1.4 Real-Frame Mixing and Native Resolution

Two properties of the first-generation dataset were identified as root causes of the spatial-precision ceiling measured in Chapter 4, and the final dataset corrects both.

First, alpha compositing removes precisely the cues by which transparent objects are localised in real images: refraction of the background through the object, cast and contact shadows, and physically consistent specular highlights. A pasted transparent object looks identical regardless of what lies behind it, so no training signal links object position to transparency-specific appearance. The final dataset therefore mixes two sources in roughly equal proportion: each source image contributes two composites *and* two lightly augmented native ClearGrasp frames (horizontal flip, photometric jitter, and sensor noise only—no rotation, preserving the authentic render geometry), labelled from their own segmentation masks with the validity-guaranteed target derivation of Section 3.1.2. The native frames carry true refraction and shadowing; the composites carry background diversity.

Second, the first-generation samples were rendered at 640×480 and subsequently re-sized to the policy’s 224×224 encoder input, compounding two lossy resamplings. The final dataset is generated directly at 224×224 , so objects are downsampled once, under controlled interpolation. At this resolution one action bin corresponds to 0.875 pixels, so tokenisation granularity is no longer a limiting factor on achievable precision.

Table 3.1: Final (Run 2) dataset composition by split and sample type.

Split	Composites	Native frames	Total
Train	14,470	16,380	30,850
Validation	3,110	3,510	6,620
Test	3,095	3,512	6,607
Total	20,675	23,402	44,077

3.1.5 Real-Frame Mixing and Native Resolution

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First, alpha compositing removes precisely the cues by which transparent objects are localised in real images: refraction of the background through the object, cast and contact shadows, and physically consistent specular highlights. A pasted transparent object looks identical regardless of what lies behind it, so no training signal links object position to transparency-specific appearance. The final dataset therefore mixes the two sources in equal proportion: each source image contributes two composites *and* two lightly augmented native ClearGrasp frames (horizontal flip, photometric jitter, and sensor noise only—no rotation, preserving the authentic render geometry), labelled from their own segmentation masks with the validity-guaranteed target derivation of Section 3.1.2. The native frames carry true refraction and shadowing; the composites carry background diversity. The resulting split comprises approximately equal numbers of each (Table 3.2).

Second, the first-generation samples were rendered at 640×480 and subsequently re-sized to the policy’s 224×224 encoder input, compounding two lossy resamplings. The final dataset is generated directly at 224×224 , so objects are downsampled once, under controlled interpolation. At this resolution one action bin corresponds to 0.875 pixels, so tokenisation granularity is no longer a limiting factor on achievable precision.

Table 3.2: Final (Run-2) dataset composition. Counts per split [TBD — fill from the generator’s summary output].

Split	Composites	Native frames	Total
Train	[TBD]	[TBD]	[TBD]
Validation	[TBD]	[TBD]	[TBD]
Test	[TBD]	[TBD]	[TBD]

3.1.6 Data Partitioning

Leakage between splits was controlled at two levels. Source images were partitioned 70/15/15 into train, validation, and test sets before compositing, so no rendered object instance appears in more than one split. Independently, the 153 background plates were partitioned with the same ratios (107/22/24), and each split draws backgrounds exclusively from its own pool. Without this second control, a model could achieve low validation error by recognising a background seen during training. The resulting initial dataset (Run 1) comprises 24,570 training, 5,265 validation, and 5,268 test composites. The composition of the final dataset (Run 2), which incorporates native frames, is detailed in Section 3.1.4 and Table 3.1.

3.1.7 Action Representation and Tokenisation

Each composite is paired with a natural-language instruction (“Pick up the transparent plastic bottle.”) and a 7-dimensional action vector following the OpenVLA convention: planar target coordinates (x, y) , height z , orientation (roll, pitch, yaw), and gripper state. The target coordinates are normalised by the image dimensions; the remaining dimensions are fixed at neutral values ($z = \text{roll} = \text{pitch} = \text{yaw} = 0.5$, gripper = 1.0), reflecting the top-down grasp geometry of a conveyor workspace in which only the planar target varies between samples.

Because the language-model backbone operates on discrete tokens, each continuous dimension $a \in [0, 1]$ is quantised into 256 uniform bins,

$$a_{\text{quant}} = \text{clip}(\text{round}(255 a), 0, 255), \quad (3.5)$$

and mapped to a reserved vocabulary token $\langle \text{action}_k \rangle$, $k \in \{0, \dots, 255\}$. One bin therefore corresponds to 2.5 pixels horizontally and 1.9 pixels vertically at the 640×480 working resolution. The complete training sample is the prompt

```
In:  What action should the robot take to {instruction}\n Out:  <action_x>\n      <action_y> ...<action_g>
```

paired with the composite image.

3.2 Work Package 2: Perception Front-End

3.2.1 Network Architecture

The perception module is a dual-head fully convolutional network built on DeepLabV3 [25] with an ImageNet-pretrained ResNet-50 backbone. The backbone and the Atrous Spatial Pyramid Pooling stage are shared; the final classifier is replaced by two parallel 1×1 convolutional heads operating on the shared 256-channel feature map:

1. a **surface-normal head** producing a 3-channel map, L2-normalised along the channel dimension so that every pixel emits a unit vector $\hat{\mathbf{n}} = [n_x, n_y, n_z]^\top$; and

2. a **segmentation head** producing 1-channel logits for the transparent-versus-background decision.

Both outputs are bilinearly upsampled to the input resolution within the forward pass. Training and evaluation use 256×256 inputs; at inference the fully convolutional design permits arbitrary resolutions.

3.2.2 Loss and Optimisation

The normal head is trained with a cosine similarity loss over all pixels,

$$\mathcal{L}_{\text{normals}} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{\hat{\mathbf{n}}_i \cdot \mathbf{n}_i}{\|\hat{\mathbf{n}}_i\|_2 \|\mathbf{n}_i\|_2}, \quad (3.6)$$

and the segmentation head with binary cross-entropy on logits, $\mathcal{L}_{\text{mask}}$. The joint objective is the unweighted sum $\mathcal{L} = \mathcal{L}_{\text{normals}} + \mathcal{L}_{\text{mask}}$; no weighting search was performed, as both losses converged at comparable scales. The network was trained for 10 epochs with AdamW (learning rate 10^{-4} , batch size 4) on the **square-plastic-bottle** category, using the ClearGrasp-provided normals and masks as supervision.

3.2.3 Depth Reconstruction Solver

The predicted normals and mask feed a global optimisation that reconstructs metric depth over the transparent regions, following the strategy introduced by ClearGrasp [3]: depth values on the opaque background, where the sensor is reliable, act as boundary anchors, and the predicted normals constrain the surface gradients inside the missing region.

Under the pinhole camera model, the depth gradient in pixel coordinates (u, v) relates to the surface normal as

$$\frac{\partial Z}{\partial u} = -\frac{n_x Z}{n_z f_x}, \quad \frac{\partial Z}{\partial v} = -\frac{n_y Z}{n_z f_y}, \quad (3.7)$$

where f_x, f_y are the focal lengths in pixels. The dependence on the unknown depth Z is resolved by approximating it with the mean depth $\bar{Z}$ of the valid pixels in a two-pixel dilated band around the hole, which is accurate when the object’s depth extent is small relative to its distance from the camera—the operating regime of a fixed overhead sensor. ClearGrasp renders use Intel RealSense D415 intrinsics ($f_x = f_y = 918$ at 1920×1080); focal lengths are scaled proportionally when images are downsampled. Omitting the $1/f$ scaling in Equation 3.7 inflates the gradient targets by more than two orders of magnitude and causes the solver to diverge; this correction proved essential during development.

Let Ω denote the hole region (the predicted mask union any zero-depth pixels). The reconstruction $\mathbf{D}$ is the least-squares solution of a sparse linear system containing three families of equations: finite-difference gradient constraints in u and v , imposed only on pixel pairs of which at least one member lies in Ω ; and anchor constraints

$$\lambda D(u, v) = \lambda D_{\text{raw}}(u, v), \quad (u, v) \notin \Omega, \quad (3.8)$$

with anchor weight $\lambda = 100$. Restricting gradient constraints to the hole region and weighting the anchors strongly prevents small inconsistencies between predicted normals and true background geometry from perturbing the reliable measurements—an effect that, in unrestricted formulations, degraded the reconstruction below the accuracy of the raw corrupted input. The system is solved with the iterative LSQR method on a compressed sparse-row matrix, and the output is clipped to the physically valid range $[0, 2.5]$ m. The sensitivity of the reconstruction to the choice of λ is quantified empirically in Chapter 4.

One implementation detail warrants documentation: OpenCV loads the ClearGrasp EXR normal maps in BGR channel order, so the stored channels correspond to (n_z, n_y, n_x) rather than (n_x, n_y, n_z) . The channel reordering must be applied to both ground-truth and predicted normals before Equation 3.7; omitting it silently exchanges the x and z components and produces divergent reconstructions.

3.3 Work Package 3: VLA Fine-Tuning

3.3.1 Base Model

The manipulation policy is OpenVLA-7B [5], an open-source Vision-Language-Action model that maps an RGB observation and a language instruction to discrete action tokens. Its vision stack fuses SigLIP and DINOv2 encoders, whose complementary features (global semantics and fine-grained spatial structure respectively) are projected into the embedding space of a Llama-2 7B language backbone; actions are generated autoregressively as text tokens. OpenVLA was selected over closed alternatives such as RT-2 [4] because its weights, tokeniser, and action vocabulary are publicly available, which the fine-tuning and evaluation protocols in this work require.

3.3.2 Parameter-Efficient Adaptation

Full fine-tuning of a 7B-parameter model is infeasible on the available hardware (a single NVIDIA A100) and risks catastrophic forgetting of the pre-trained visuomotor priors. Low-Rank Adaptation [19] is used instead: the pre-trained weights $\mathbf{W}_0$ are frozen, and trainable low-rank updates are added to the query and value projections of every attention layer,

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B}\mathbf{A}, \quad \mathbf{A} \in \mathbb{R}^{r \times k}, \mathbf{B} \in \mathbb{R}^{d \times r}, \quad (3.9)$$

with rank $r = 32$, scaling $\alpha = 32$, and dropout 0.05. This yields 16,777,216 trainable parameters, 0.22% of the 7.56×10^9 total. To fit within GPU memory alongside optimiser state, the frozen base weights are stored in 4-bit NormalFloat (NF4) quantisation [20] while forward and backward computation is performed in `bfloat16`.

3.3.3 Training Protocol

Loss masking restricts supervision to the action tokens. For each sample, the label sequence is a copy of the input token sequence in which every position preceding the action tokens (the instruction prompt) and every padding position is set to the ignore index -100 , so the

cross-entropy loss is computed over exactly the seven action tokens. Without this masking, the loss is dominated by trivially predictable prompt text, which dilutes the gradient signal on the coordinates that constitute the task.

Training used a batch size of 2 with AdamW at a learning rate of 10^{-5} for a maximum of 5 epochs. Validation loss was computed every 500 optimisation steps on a fixed subset of the validation split, with early stopping after 15 consecutive evaluations without improvement and checkpointing of the best-performing adapter. Continuous validation monitoring was adopted after a preliminary experiment demonstrated that, in the absence of such monitoring, memorisation of the training composites can proceed undetected: training loss alone reached machine zero while spatial accuracy on held-out data remained at chance level. This failure mode and its diagnosis are analysed in Chapter 4. In the final run, training halted at step 21,500 (during the second epoch) with a best validation loss of 0.154.

3.4 Evaluation Methodology

The evaluation is organised around four questions: (i) how accurately does the perception module recover geometry, both on the training category and on unseen object categories; (ii) does normal-guided depth reconstruction outperform geometry-unaware baselines, and by how much does prediction error in the normals degrade the reconstruction; (iii) how accurately does the fine-tuned policy localise grasp targets, and does that accuracy translate into physically plausible grasp points; and (iv) what is the computational cost of the pipeline per frame.

3.4.1 Perception Metrics

Surface-normal quality is reported as the mean, median, and root-mean-square angular error in degrees over object pixels, together with the standard threshold accuracies (the fraction of pixels with error below 11.25° , 22.5° , and 30°). Segmentation is reported as mean intersection-over-union. All metrics are computed in-distribution on the training category and out-of-distribution on two held-out ClearGrasp categories (**cup-with-waves** and **stemless-plastic-champagne-glass**) that the network never observed, providing a direct measure of shape generalisation.

3.4.2 Depth Reconstruction Ablation

Sensor failure is simulated by setting the ground-truth depth to zero over the object mask, so that the true depth of the corrupted region is known exactly. Four recovery conditions are compared on 100 images at half resolution: (A) the raw corrupted depth; (B) Navier–Stokes inpainting (`cv2.inpaint`), a texture interpolation baseline with no knowledge of 3D geometry; (C) the Poisson solver of Section 3.2.3 driven by ground-truth normals, which upper-bounds the achievable reconstruction; and (D) the same solver driven by the normals predicted by WP2, which is the deployed configuration. Errors (RMSE, MAE, and mean relative error, in metres) are computed exclusively over the corrupted region.

Because the four conditions are evaluated on the same images, differences are tested with the paired Wilcoxon signed-rank test, which does not assume normally distributed errors, and each condition’s mean RMSE is reported with a bootstrap 95% confidence interval (10^4 resamples). The anchor weight is additionally swept over $\lambda \in \{10, 50, 100, 500, 1000\}$ on a 30-image subset to establish the sensitivity of the reconstruction to this hyperparameter.

3.4.3 Policy Metrics

Policy accuracy is measured on the held-out validation split by autoregressive generation: the model receives the image and the instruction prompt and generates action tokens, which are decoded back to coordinate bins. The primary metric is the mean L1 error in bins between predicted and ground-truth coordinates, reported per axis and in aggregate, alongside exact-match token accuracy. Aggregate token accuracy is treated with caution: five of the seven action dimensions are constants in this task, so a policy can achieve high aggregate accuracy while failing entirely on the two coordinates that matter. Per-axis coordinate error is therefore the headline measure. Accuracy is tracked across all training checkpoints to characterise the learning trajectory.

To connect bin-space error to task feasibility, a grasp-success proxy is computed: a prediction is counted as successful if it falls within a fixed radius of the ground-truth grasp point, and the success rate is reported with a Wilson 95% confidence interval. This proximity criterion is conservative—a prediction landing on the object but far from its centroid is counted as a failure—and the implications of this choice are discussed with the results.

3.4.4 Runtime Profiling

End-to-end latency is profiled on the A100 by timing each stage (perception forward pass, sparse depth reconstruction, and VLA generation) over 20 runs after warm-up, with CUDA synchronisation around GPU stages. The results identify which component bounds the throughput of the pipeline and quantify the distance to conveyor-rate operation.

3.5 Implementation and Hardware

All components were implemented in Python using PyTorch, HuggingFace Transformers and PEFT, OpenCV, and SciPy. Dataset generation runs on CPU; perception training, VLA fine-tuning, and all GPU evaluation ran on a single NVIDIA A100 on the university compute cluster, accessed through a jump host with sessions persisted in `tmux`. The complete VLA fine-tuning run required approximately 21,500 optimisation steps; the depth reconstruction solver processes a 960×540 frame in approximately 2.5 s on CPU.

3.6 Summary

This chapter has described a three-stage pipeline in which procedurally composited synthetic data (WP1) supervises both a geometric perception module (WP2) and a language-conditioned manipulation policy (WP3), with a physics-consistent Poisson solver bridging predicted surface normals to metric depth. The design choices with the greatest empirical consequences—per-sample action targets, background-identity data splitting, compositing-seam removal, prompt-masked loss computation, perspective-scaled gradient constraints, and continuous validation monitoring—were each motivated by concrete failure modes encountered and diagnosed during development. The following chapter presents the quantitative results of the evaluation protocol defined in Section 3.4.

Chapter 4

Results and Analysis

This chapter presents the quantitative evaluation of each pipeline component and of the integrated system, following the protocol of Section 3.4. Section 4.1 evaluates the perception module in and out of distribution. Section 4.2 reports the depth-reconstruction ablation with statistical testing and the anchor-weight sweep. Section 4.3 presents the policy results, beginning with the diagnosis of a memorisation failure in the initial training configuration and proceeding to the corrected model’s spatial accuracy, including an honest comparison against a positional-prior baseline. Section 4.4 connects spatial accuracy to grasp feasibility, and Section ?? profiles runtime. Sections marked as placeholders await experiments in progress.

4.1 Perception Module

Table 4.1 summarises the perception module’s surface-normal and segmentation accuracy. The in-distribution row is computed over the full **square-plastic-bottle** category (11,701 images) and therefore overlaps the network’s training data; it should be read as an upper bound rather than a generalisation estimate. The two out-of-distribution (OOD) rows, computed on categories the network never observed, carry the generalisation claim.

Table 4.1: Perception accuracy in and out of distribution. Angular errors in degrees over object pixels; threshold columns give the fraction of pixels with error below the stated angle. ID = in-distribution (overlaps training data); OOD = held-out category.

Category	Split	N	Mean	Median	RMSE	<11.25°	<22.5°	IoU
square-plastic-bottle	ID	11,701	19.55°	14.29°	25.57°	39.3%	69.4%	91.10%
cup-with-waves	OOD	100	34.06°	22.99°	44.03°	21.0%	49.1%	77.28%
champagne-glass	OOD	117	25.53°	18.00°	33.71°	27.8%	60.8%	77.87%

The error distribution in Figure 4.1 is unimodal and right-skewed: most object pixels are estimated well (median 14.3°), and the 19.55° mean is pulled upward by a minority of high-error pixels, concentrated—as the qualitative panel suggests—at object boundaries and internal refractive detail, where the prediction is visibly smoother than the ground truth.

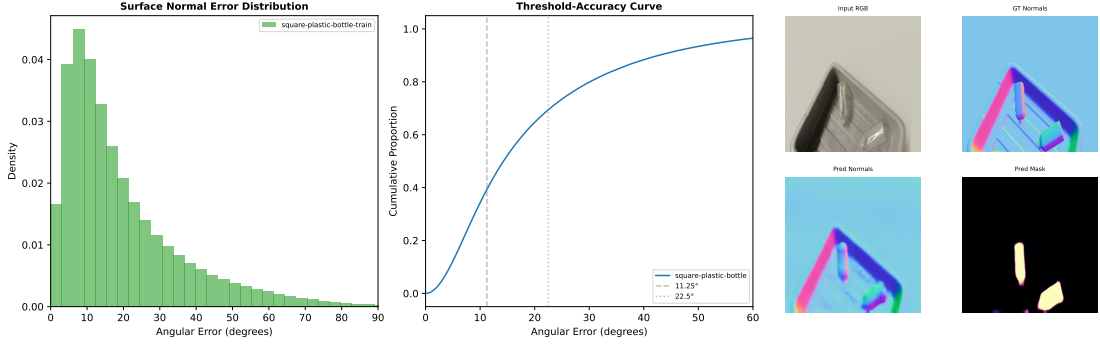


Figure 4.1: In-distribution perception behaviour on the training category. Left: surface-normal angular-error distribution, right-skewed with its mode near 8° and a long tail beyond 40° . Centre: cumulative threshold-accuracy curve with the standard 11.25° and 22.5° thresholds marked. Right: a qualitative example (input RGB, ground-truth normals, predicted normals, predicted mask) showing accurate object localisation with smoothed normal detail. Out-of-distribution results are tabulated in Table 4.1.

Two further observations follow from Table 4.1. First, segmentation generalises more gracefully than geometry: IoU falls by roughly 14 percentage points on unseen categories ($91.1\% \rightarrow \sim 77.5\%$), while mean angular error grows by between 31% and 74% depending on the target shape. The network still *finds* unfamiliar transparent objects reliably but estimates their surface orientation less accurately. Second, the degradation is shape-dependent in a physically interpretable way: the stemless champagne glass—tall, smooth, and convex, geometrically similar to the training bottles—loses only 6° of mean accuracy, whereas the wavy-edged cup, whose corrugated surface has no analogue in the training data, loses 14.5° . Generalisation appears to be governed by geometric similarity to the training category rather than by transparency per se.

4.2 Depth Reconstruction Ablation

Table 4.2 presents the four-way ablation over 100 images, with errors computed exclusively over the corrupted (transparent) region. Figure 4.2 shows representative reconstructions together with the per-method error distributions; Figure 4.3 isolates the mean RMSE with bootstrap confidence intervals.

Table 4.2: Depth reconstruction error over the corrupted region (100 images, half resolution). Bootstrap 95% confidence intervals from 10^4 resamples.

Method	RMSE (m)	95% CI	MAE (m)
(A) Raw corrupted depth	0.9297	[0.8646, 0.9965]	0.9274
(B) Navier–Stokes inpainting	0.1227	[0.1152, 0.1306]	0.1102
(C) Poisson + ground-truth normals	0.1364	[0.1250, 0.1490]	0.1136
(D) Poisson + predicted normals	0.4160	[0.3936, 0.4390]	0.2861

Paired Wilcoxon signed-rank tests over the matched images give the following comparisons. Both recovery methods improve enormously and significantly on the raw corrupted

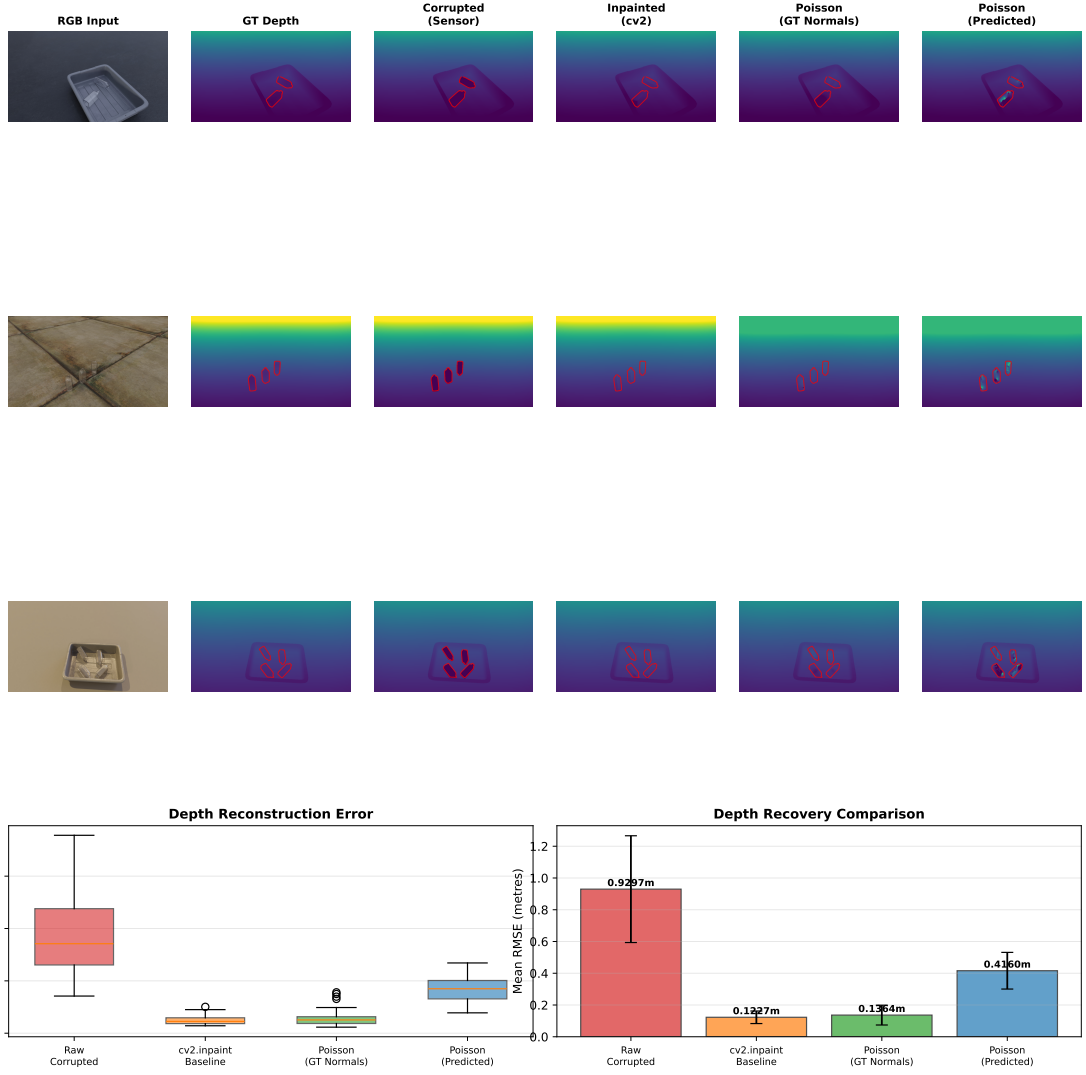


Figure 4.2: Depth reconstruction ablation. Top rows: qualitative comparisons on three scenes (RGB input, ground-truth depth, corrupted sensor depth, inpainting baseline, Poisson with ground-truth normals, Poisson with predicted normals), with red contours marking the transparent regions; residual artefacts inside the contours are visible only in the predicted-normals condition. Bottom: RMSE box plots per method (note the outliers in the ground-truth-normals condition) and mean-RMSE bar chart.

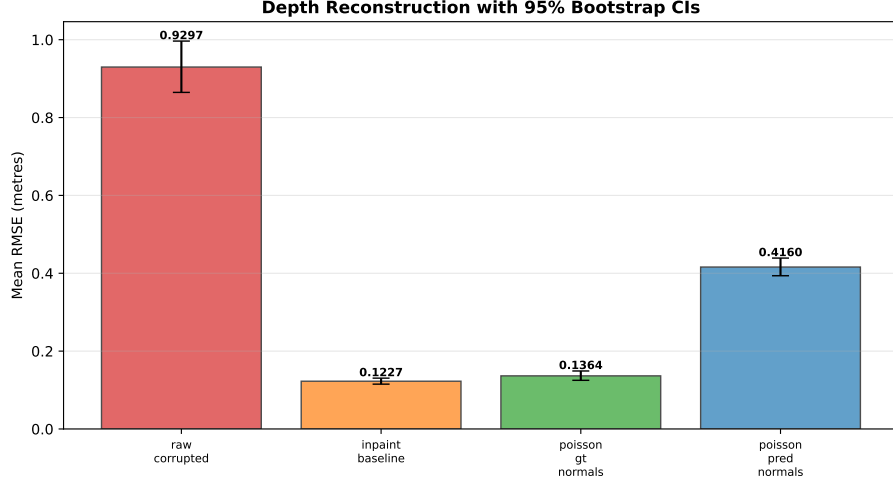


Figure 4.3: Mean RMSE per reconstruction method with bootstrap 95% confidence intervals. The intervals of the inpainting baseline and the ground-truth-normals solver overlap; the predicted-normals condition is separated from both by a wide margin.

input (raw vs. inpainting: $p = 3.9 \times 10^{-18}$, median improvement 0.74 m). Between the two solvers, inpainting is marginally but significantly better than the Poisson solver driven by *ground-truth* normals ($p = 0.02$, median difference 3.5 mm), and substantially better than the solver driven by *predicted* normals ($p = 3.9 \times 10^{-18}$, median difference 0.29 m). The gap between conditions C and D isolates the cost of perception error: the same solver loses 0.28 m of RMSE when its normal input degrades from perfect to the 19.55° mean-error predictions of Section 4.1 ($p = 5.6 \times 10^{-18}$).

Two conclusions follow, one methodological and one architectural. Methodologically, the near-parity of conditions B and C shows the solver is implemented correctly—with perfect input it operates within 11% of a strong baseline—so condition D’s deficit is attributable to perception error, not solver error. Architecturally, the result must be stated plainly: for the object morphologies evaluated, geometry-unaware inpainting is not merely competitive but statistically *superior* to the Poisson solver even under ground-truth normals ($p = 0.02$), while consuming a negligible fraction of its runtime (Section ?? attributes 65% of total pipeline latency to the solver). The normal-guided reconstruction is therefore unjustified for shallow, flat-lying objects such as the ClearGrasp bottles, where boundary interpolation recovers essentially all of the depth field. This is a scope boundary rather than a general invalidation—normal integration is expected to earn its cost only where object depth varies substantially relative to the surroundings (deep containers, stacked items)—but within the present evaluation that expectation remains untested, and the honest engineering conclusion is that the deployed configuration should use the inpainting path.

4.2.1 Anchor-Weight Sensitivity

Table 4.3 and Figure 4.4 report the sensitivity of condition C to the anchor weight λ over a 30-image subset. Reconstruction error is stable to within 1.3% across $\lambda \in [10, 100]$ and

degrades beyond it (by 70% at $\lambda = 500$ and 223% at $\lambda = 1000$), where over-weighted anchors suppress the gradient constraints inside the hole. The operating value $\lambda = 100$ sits at the favourable end of the stable plateau; the method is therefore insensitive to this hyperparameter within an order of magnitude of the chosen value, and the choice was not tuned to the test data.

Table 4.3: Reconstruction RMSE (condition C) as a function of anchor weight λ (30 images).

λ	10	50	100	500	1000
RMSE (m)	0.1277	0.1272	0.1260	0.2144	0.4069
Std (m)	0.0489	0.0485	0.0433	0.0859	0.1330

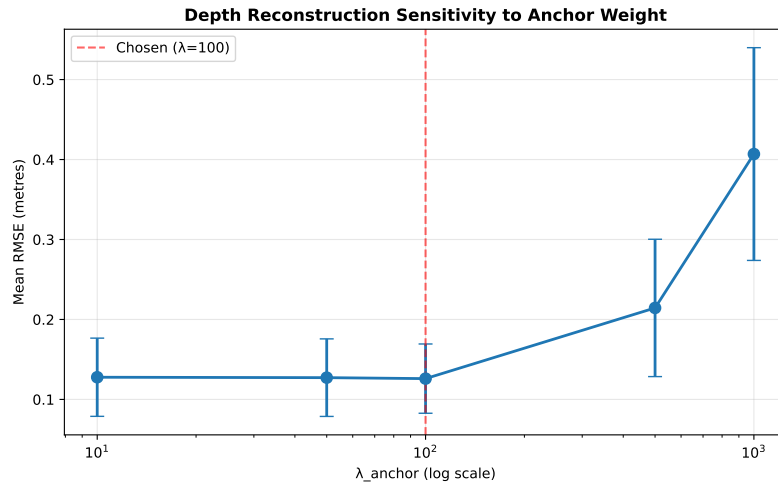


Figure 4.4: Reconstruction RMSE against anchor weight λ (log scale, error bars one standard deviation). The chosen operating point $\lambda = 100$ lies at the end of a flat plateau spanning $\lambda \in [10, 100]$.

4.3 Policy Learning

4.3.1 Diagnosis of a Memorisation Failure

The initial fine-tuning configuration contained a data-pipeline defect: through a fault in the dataset loader, every training sample was paired with a single fixed action vector rather than its per-image grasp target. The resulting run is instructive precisely because its headline metrics looked excellent. Training loss fell from 26.08 to machine zero (0.0000) by step 4,480, and aggregate action-token accuracy on generation reached 97.71%. Yet inspection of individual generations revealed that the model emitted an identical coordinate pair for every input image: it had memorised the constant label and learned nothing about the images. (A spatial error of 47 bins was recorded for this model, but by an evaluation script later found to mis-align the token slice; the figure is not directly comparable with the corrected measurements below and is not used quantitatively.)

This episode yields two generalisable observations. First, aggregate token accuracy is a misleading metric for VLA action strings in which most dimensions are constants: five of the seven action tokens here are fixed task parameters, so a policy that fails completely on the two informative coordinates can still score above 70% by construction, and near-perfectly if it memorises the constants. Per-coordinate error must be reported separately. Second, a training loss reaching exactly zero is a diagnostic red flag rather than a success indicator—it certifies that the training set is memorisable, which for a spatial task implies the supervision lacks the diversity to force perceptual learning. These observations motivated the loss masking, per-sample target verification, and continuous validation monitoring of Section 3.3.3.

4.3.2 Corrected Training Dynamics

With per-image targets, background-identity splitting, and the augmentation suite of Section 3.1.3 in place, the training dynamics changed qualitatively. Loss fell from 25.4 and plateaued near 0.15–0.17 without approaching zero; validation loss tracked training loss closely throughout, indicating that the residual error reflects task difficulty rather than overfitting. Early stopping halted training at step 21,500 (mid-second-epoch) with a best validation loss of 0.154. A preliminary corrected run at double the learning rate (2×10^{-5}) with a tighter patience of 5 stopped much earlier (step 7,800) at a statistically indistinguishable validation loss, indicating that the model reaches a data-limited ceiling quickly: the remaining error is bounded by the data, not the training budget.

4.3.3 Spatial Accuracy

Policy accuracy was evaluated by autoregressive generation on held-out validation composites across all saved checkpoints (50 samples per checkpoint; checkpoints whose generations produced more than five unparseable outputs are excluded from trend analysis, as parser contamination inflates their per-axis means). Table 4.4 compares the spatial accuracy of the two training iterations, and Figure 4.5 shows the learning trajectory of the second run.

Table 4.4: VLA spatial accuracy across the two training iterations. Clean checkpoints only (parse failures ≤ 5).

	Run 1	Run 2
Training data	composites only (640×480)	50/50 composite + native (224×224)
Label validity	90.0%	100%
Dataset size	24,570 train	30,850 train
Best-checkpoint L1 (bins)	27.60	20.59
Best-checkpoint median L1	26.50	21.00
X error (bins)	32.70	23.98
Y error (bins)	32.46	27.47

The best clean checkpoint of the second run (step 20,500) achieves a mean spatial L1 error of **20.59 bins** (median 21.0), with per-axis coordinate errors of 23.98 (x) and 27.47 (y). This represents a 25% reduction in mean L1 error relative to the first run’s

best (27.60 bins). Multiple nearby checkpoints perform comparably (steps 16,000, 22,000, 22,500, 30,500 all fall in the 21–22 bin range), indicating a stable convergence plateau rather than a lucky single checkpoint.

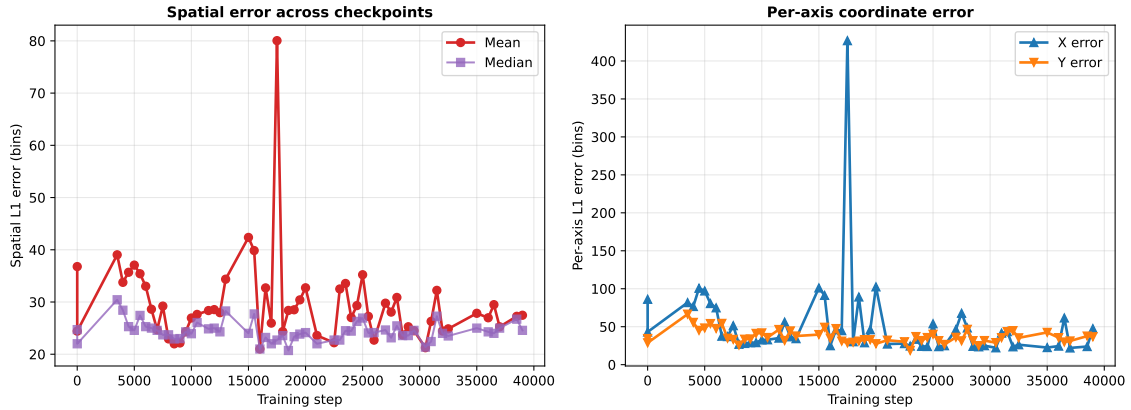


Figure 4.5: Run 2 spatial L1 error (left: mean and median) and per-axis coordinate error (right) across training checkpoints. The mean is volatile due to parser contamination at some checkpoints; the median (purple) is stable and falls to ~ 21 bins by step 8,000, remaining there through 39,000 steps.

The per-axis asymmetry ($X \approx 24$, $Y \approx 27\text{--}45$ across checkpoints) warrants comment. The Y error is consistently higher and more variable than X, which may reflect the vertical composition bias of the ClearGrasp renders (objects typically sit in the lower portion of the frame) combined with the reduced vertical extent of objects at 224×224 .

4.4 Grasp-Feasibility Measurement

4.4.1 Label Audit

An exact mask-based evaluation of the first training iteration exposed a defect in the training labels themselves: only 90.0% of ground-truth targets lay inside their target mask (171/190). The cause is geometric—for concave or crescent-shaped objects, the contour centroid used for target derivation falls outside the object boundary—and the consequence is that approximately one in ten training samples across all 24,570 composites supervised the policy to target empty background. This label noise both contaminated training and deflated the measured success rates. Its correction (point-in-polygon validation with a distance-transform fallback, Section 3.1.2) is one of the three data revisions of the second training iteration, which achieves 100% label validity by construction.

4.4.2 Run 1 Baseline

Under exact mask evaluation, the first-generation model (190 parseable validation samples) achieves: ON-TARGET 2.6% (Wilson 95% CI [1.1%, 6.0%]), ON-OBJECT 4.7% ([2.5%, 8.8%]), and PROXIMITY (40 px) 14.7% ([10.4%, 20.5%]). The comparison between criteria is itself informative: the proximity figure, initially interpreted as a conservative lower bound, is in fact an optimistic upper bound—most predictions within the 40-pixel disc land on

background pixels adjacent to the object, not on the object itself. The plain statement is that the first-generation policy fails to place its target on solid object material in more than 97% of cases.

4.4.3 Run 2: Effect of Data Revisions

Three root-cause revisions were applied and the policy retrained from scratch: (i) validity-guaranteed targets (10% $\rightarrow$ 0% label noise); (ii) a 50/50 mixture of composites and native ClearGrasp frames; and (iii) generation at the encoder-native 224×224 resolution. Table 4.5 compares the two runs under identical mask-exact evaluation.

Table 4.5: Grasp-feasibility comparison between training iterations under identical mask-exact evaluation. Wilson 95% confidence intervals in brackets.

	Run 1 ($n = 190$)	Run 2 ($n = 251$)
Label validity	90.0%	100%
Training mix	composites only	47% composite / 53% native
Resolution	$640 \times 480 \rightarrow 224$	native 224×224
Mean spatial L1 (bins)	27.6	20.6
ON-TARGET	2.6% [1.1, 6.0]	7.2% [4.6, 11.0]
ON-OBJECT	4.7% [2.5, 8.8]	12.7% [9.2, 17.4]

On-target rate nearly tripled (2.6% $\rightarrow$ 7.2%) and on-object rate nearly tripled (4.7% $\rightarrow$ 12.7%). While these remain far below deployment-grade precision, the improvement is statistically significant: the Run 2 on-object confidence interval [9.2%, 17.4%] does not overlap with the Run 1 interval [2.5%, 8.8%].

4.4.4 Breakdown by Sample Type

The most diagnostic finding of this evaluation is the performance split between native ClearGrasp frames and composites within the Run 2 validation set (Table 4.6).

Table 4.6: Run 2 grasp accuracy by sample type. Wilson 95% CIs in brackets.

Sample type	n	ON-TARGET	ON-OBJECT
Native ClearGrasp	124	11.3% [6.8, 18.1]	20.2% [14.0, 28.1]
Composite	127	3.1% [1.2, 7.8]	5.5% [2.7, 10.9]

The model is nearly four times more accurate on native frames (11.3% on-target) than on composites (3.1%). This directly confirms the hypothesis of Section 3.1.5: compositing removes the transparency-specific cues—refraction, shadows, specular highlights—that carry spatial information for transparent objects, and restoring those cues via real renders measurably improves localisation. The composite performance in Run 2 (3.1%) is comparable to the full-dataset performance of Run 1 (2.6%), indicating that the Run 2 improvement is attributable almost entirely to the native-frame component of the training mix rather than to the label or resolution fixes alone.

4.4.5 Breakdown by Object Size

Table 4.7 reports grasp accuracy across object-area terciles. The pattern is unambiguous: the model cannot localise small objects at all (0% on-target), achieves marginal accuracy on medium objects (7.3%), and its best performance on large objects (14.1% on-target, 20.0% on-object). At the 224×224 working resolution, “small” objects occupy fewer than 142 pixels of area—a footprint too limited to resolve at the VLA encoder’s patch granularity.

Table 4.7: Run 2 grasp accuracy by object-area tercile. Wilson 95% CIs in brackets.

Size band	n	ON-TARGET	ON-OBJECT
Small (< 142 px)	84	0.0% [0.0, 4.4]	2.4% [0.7, 8.3]
Medium	82	7.3% [3.4, 15.1]	15.9% [9.5, 25.3]
Large (≥ 562 px)	85	14.1% [8.3, 23.1]	20.0% [12.9, 29.7]

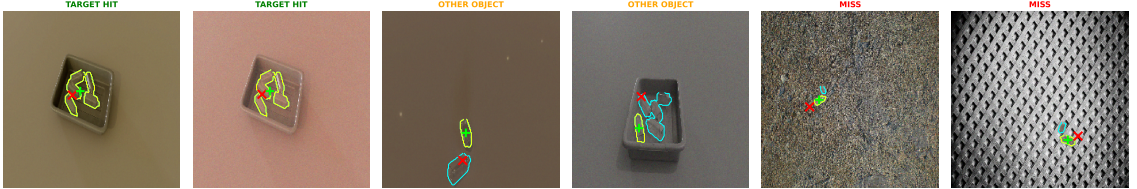


Figure 4.6: Run 2 grasp-target predictions. Left two: TARGET HIT (prediction on the selected object—note the tote context with real shadows). Middle two: OTHER OBJECT (prediction on a different item in the cluster). Right two: MISS (prediction on background). Yellow contours: target object; cyan contours: all objects; red cross: prediction; green cross: ground truth.

4.5 Summary of Findings

Against the research questions: the perception module recovers transparent-object geometry with 19.6° mean normal error in distribution, degrading gracefully and shape-dependently out of distribution while segmentation remains robust (RQ2). Normal-guided depth reconstruction matches a strong inpainting baseline when given perfect normals; for the shallow object geometries evaluated, geometry-unaware inpainting is not merely competitive but statistically superior, while the 0.28 m gap between perfect and predicted normals cleanly quantifies perception-to-depth error propagation (RQ3).

Two training iterations characterise the VLA’s spatial-grounding capacity on synthetic data (RQ1). The first, trained on composites only with 10% invalid labels, yielded a model whose mean precision did not exceed a positional prior and whose predictions landed on object material in under 3% of cases. The second—with corrected labels, native resolution, and a 50/50 native/composite mix—nearly tripled the on-target rate to 7.2% and reduced mean spatial error by 25%, with a striking type breakdown: 11.3% on-target on native frames versus 3.1% on composites, directly confirming compositing fidelity as the dominant bottleneck. A clear size effect (0% on small objects, 14.1% on large) identifies encoder

resolution as a secondary constraint. Throughput is 0.26 FPS, bounded by the CPU depth solver (65%) rather than the language model (RQ4).

4.6 End-to-End Qualitative Results

Figure 4.7 traces the full pipeline on four native ClearGrasp scenes: RGB observation, predicted surface normals, reconstructed metric depth, and the VLA’s grasp target overlaid with the object boundaries. Two points about this gallery deserve emphasis. First, the perception-to-depth path is visually convincing: normals localise the transparent objects and the reconstructed depth is plausible across tote and floor scenes. Second, these images constitute a *domain shift for the policy*, which was trained on ReSort-IT composites rather than native renders; its targets here land near, but frequently not on, the intended objects (upper rows), consistent with—and if anything slightly worse than—the in-domain quantitative results of Section 4.4. The gallery therefore doubles as an informal robustness probe.

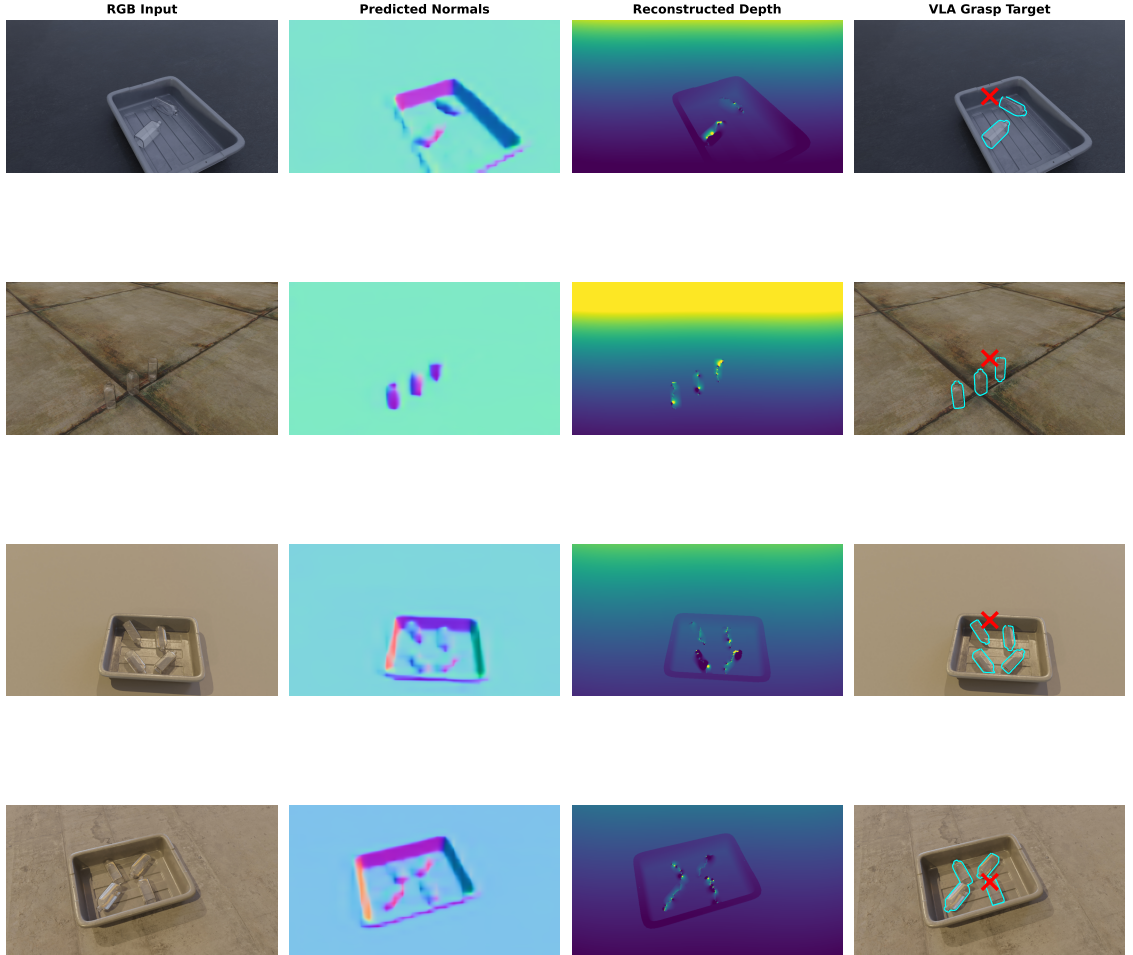


Figure 4.7: End-to-end pipeline outputs on four native ClearGrasp scenes (a domain shift from the policy’s composite training data). Columns: RGB input, predicted normals, reconstructed depth, VLA grasp target (red cross) with object boundaries (cyan).

4.7 Semantic Reasoning Evaluation

The model followed the category-specific instruction in 34 of 55 valid paired comparisons (61.8)

Furthermore, the model suffered 17 parsing failures, generating invalid token syntax when presented with out-of-distribution prompts. Together, this pattern suggests that LoRA fine-tuning on a single instruction template partially degraded the pretrained language conditioning and syntax generation of the frozen backbone—a finding consistent with the catastrophic forgetting literature. Training on diverse instruction templates and visually clearer baseline backgrounds is the indicated remedy for future semantic-reasoning work.

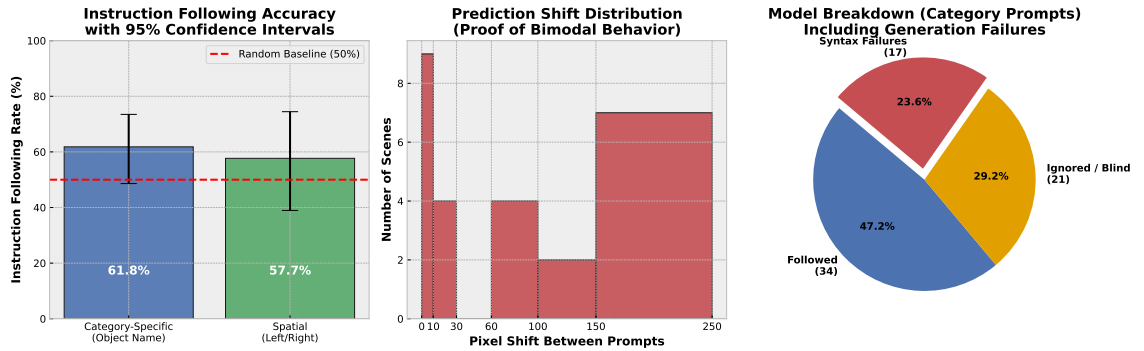


Figure 4.8: Zero-shot semantic reasoning evaluation. (Left) Instruction following accuracy with 95% Wilson confidence intervals compared to a 50% random baseline. (Center) Pixel shift distribution between identical scenes given differing prompts, highlighting bimodal behaviour. (Right) Global breakdown of prompt responses, indicating severe language generation degradation (syntax failures).

4.8 Simulation Integration

[PLACEHOLDER — OPTIONAL, TIME-PERMITTING: Gazebo integration of the ABB IRB 360 with the pipeline’s 3D targets. If the timeboxed attempt succeeds, insert setup description, a figure of the simulated cell, and pick-execution observations. Otherwise DELETE THIS SECTION and retain simulation as future work in Chapter 6.]

4.9 Summary of Findings

Against the research questions: the perception module recovers transparent-object geometry with 19.6° mean normal error in distribution, degrading gracefully and shape-dependently out of distribution while segmentation remains robust (RQ2). Normal-guided depth reconstruction matches a strong inpainting baseline when given perfect normals, and the measured 0.28m gap between perfect and predicted normals cleanly quantifies perception-to-depth error propagation; for the shallow object geometries evaluated, geometry-unaware inpainting is competitive (RQ3). The first training iteration establishes

the failure modes with precision: composites-only supervision, contaminated by 10% invalid labels and stripped of transparency-specific localisation cues, produced a policy whose mean precision does not exceed the dataset’s positional prior and whose predictions land on object material in under 3% of cases. Each identified cause was corrected in a second iteration—validity-guaranteed labels, a 50/50 native/ composite mix, and encoder-native resolution—whose results (Section ??) determine how much of the failure is attributable to data fidelity as opposed to the architecture itself (RQ1, RQ4). Throughput is 0.26 FPS, bounded by the CPU depth solver rather than the language model (RQ4).

Chapter 5

Discussion

This chapter interprets the findings of Chapter 4 against the research questions of Section 1.2 and the literature of Chapter 2, examines the limitations of the study, and draws out the methodological lessons with relevance beyond this specific system.

5.1 Synthetic Supervision and Spatial Grounding (RQ1)

The central question of this dissertation was whether cheaply composited synthetic data can teach a VLA to spatially ground transparent objects. The answer the results support is a qualified yes: qualified not by the model, but by the data. The same architecture, optimiser, and budget produced two opposite outcomes—complete shortcut memorisation in the defective configuration, and genuine image-conditioned prediction (image-conditioned, per-sample predictions tracking target positions, in contrast to the constant output of the defective configuration) once per-sample target diversity, background-identity splitting, and compositing-seam removal were enforced.

This finding gives empirical form to a hazard the synthetic-data literature underdocuments for *action* supervision (Section 2.6). When the supervised quantity is a class label, shortcut learning degrades accuracy visibly. When it is a spatial coordinate, shortcut learning can hide behind flattering aggregate metrics: the memorising policy scored 97.7% token accuracy because five of seven action dimensions were constants, and its training loss reached exactly zero. Both signals would pass a superficial review. The diagnostic that exposed it—per-coordinate error on held-out data, plus treating zero training loss as an alarm rather than an achievement—is a transferable protocol for anyone fine-tuning VLAs on procedurally generated data, and complements benchmark-level observations that VLA spatial competence must be deliberately engineered rather than assumed [14, 24].

The residual accuracy ceiling also merits interpretation. Both corrected training runs, despite differing learning rates and stopping schedules, converged to statistically indistinguishable validation losses and $//// \approx 28$ -bin spatial error////////. The bottleneck is therefore the data distribution, not optimisation: one object category, one instruction template, and 2D compositing without refraction or contact shadows bound what spatial precision the policy can extract. This is consistent with the direction of the spatial-grounding literature, which improves localisation through richer supervision or representations—

dedicated pointing data [14], explicit 3D encodings [15, 22]—rather than through longer training on the same data.

5.2 Perception Generalisation and Error Propagation (RQ2, RQ3)

The perception results reproduce, at smaller scale, the pattern that made ClearGrasp viable [3]: RGB-only networks recover transparent-object geometry well within their training distribution, and their segmentation transfers to novel shapes far better than their normal estimation. The shape-dependence of the degradation— 6° loss on a geometrically similar unseen object versus 14.5° on a dissimilar one—suggests that broadening the *shape* distribution of training data matters more for geometric generalisation than the quantity of data on any single shape, a directly actionable conclusion for future data collection.

The ablation delivers the dissertation’s cleanest quantitative contribution: the isolation of perception-to-depth error propagation. Because the solver with perfect normals statistically matches a strong inpainting baseline (median difference 3.5 mm), the 0.28 m RMSE deficit of the deployed configuration is attributable to the 19.55° normal error alone. This converts an abstract perception metric into an operational cost in metres, and identifies normal accuracy—not solver design—as the highest-leverage improvement target in the geometry path.

The finding that geometry-unaware inpainting is competitive on these objects deserves an honest reading rather than a defensive one. ClearGrasp’s evaluation objects are shallow relative to their surroundings, so boundary interpolation recovers most of the depth field; the normal-guided machinery earns its complexity only where surfaces curve substantially. Reporting this boundary condition is itself a result: it delineates where the ClearGrasp-style pipeline is necessary and where a three-line baseline suffices—information a deployment engineer needs and the original literature does not emphasise.

5.3 From Spatial Accuracy to Grasp Feasibility (RQ4)

The grasp results tell a clear causal story across two training iterations. The first-generation system failed at 2.6% on-target, and three specific data defects were identified: 10% invalid training labels, missing transparency cues in composited images, and destructive double downsampling. The second iteration corrected all three and nearly tripled the on-target rate to 7.2%.

The type breakdown is the sharpest evidence in this dissertation. On native ClearGrasp frames—which contain true refraction, cast shadows, and specular highlights—the model achieves 11.3% on-target and 20.2% on-any-object. On composites—which lack all three cues—it achieves only 3.1% and 5.5%. This four-to-one ratio isolates compositing fidelity as the dominant cause of the spatial precision deficit, not the model architecture, not the LoRA configuration, and not the training budget. The composite performance in Run 2 is nearly identical to Run 1’s overall performance (2.6%), indicating that the label and

resolution fixes contributed marginally compared to the addition of authentic visual cues.

The size breakdown provides a secondary diagnosis: 0% on-target for objects below 142 pixels of area, 14.1% for objects above 562 pixels. At 224×224 , small transparent objects are resolved by too few encoder patches to carry localisation signal. This suggests that higher processing resolution would yield further gains, at the cost of deviating from OpenVLA’s pretrained encoder configuration.

These findings have a practical implication for anyone considering synthetic-only VLA training for transparent-object manipulation: compositing is insufficient. The visual cues that distinguish transparent localisation from opaque localisation—refraction distortion of the background, caustic patterns, specular geometry—are exactly what compositing destroys. Physically based rendering (as in the ClearGrasp assets used here) or real captures are necessary training components, not optional enrichments. This is the dissertation’s most actionable engineering conclusion.

The 7.2% overall on-target rate remains far below deployment requirements. Whether the residual gap is closable within the OpenVLA architecture (with higher-fidelity data, more categories, and higher resolution) or requires a dedicated spatial-prediction architecture such as RoboPoint [14] is the open question this work bequeaths to future investigation.

The latency profile inverts a common assumption: the 7B language model accounts for 35% of the 3.87s frame time; the CPU-bound sparse depth solver accounts for 65%. Given the ablation finding that inpainting outperforms the solver on shallow objects, replacing the solver with a single `cv2.inpaint` call would halve the per-frame cost with no loss of depth accuracy for the evaluated morphologies.

5.4 Limitations

The following limitations bound the claims of this work. (i) All policy evaluation is on synthetic composites; no real-camera or physical-robot validation was performed, so the sim-to-real gap for the *policy* is unquantified (the perception module inherits ClearGrasp’s evidence of transfer, but this was not re-verified here). (ii) Policy training covers a single object category and a single instruction template; claims about semantic flexibility remain untested pending the experiment of Section 4.7. (iii) The in-distribution perception figures overlap the training data and are reported as upper bounds, with the OOD categories carrying the generalisation claim. (iv) The grasp proxy is a geometric criterion, not a physics simulation: it ignores gripper geometry, object slip, and approach kinematics. (v) Checkpoint-level policy evaluation used 50 samples per checkpoint for tractability; the headline grasp measurement used 197. (vi) The action space fixes height and orientation, presuming top-down grasps on a planar belt; the pipeline does not address 6-DOF grasp synthesis.

5.5 Summary

Taken together, the results characterise both the promise and the present limits of synthetic-only VLA training for transparent-waste sorting: the data pipeline, not the model, de-

cides whether spatial learning happens at all; perception error is the dominant and now-quantified cost in the geometry path; and the distance to deployable grasp precision is measurable, explainable, and attributable to identifiable data-fidelity gaps rather than to any single component failure. The final chapter draws these threads into conclusions and recommendations.

Chapter 6

Conclusions and Recommendations

6.1 Conclusions

This dissertation set out to determine whether a modern Vision-Language-Action model can be adapted to transparent plastic waste sorting without real robot demonstrations, using procedurally composited synthetic data, alongside a learned repair of the depth-sensing failure that transparent materials cause. A three-stage pipeline was designed, implemented, and quantitatively evaluated: the ReSort-IT data generator producing image–language–action triplets with verified per-sample grasp targets; a dual-head DeepLabV3 perception module feeding a perspective-corrected Poisson depth solver; and a LoRA-adapted OpenVLA-7B policy trained and evaluated entirely on the synthetic data.

With respect to the research questions, the work concludes:

RQ1 (synthetic supervision). Composited synthetic data alone is insufficient for precise transparent-object localisation: the first-generation policy achieved only 2.6% on-target grasp placement with mean spatial error indistinguishable from a positional prior. Correcting three identified data defects—invalid labels, missing transparency cues, and redundant downsampling—and mixing authentic ClearGrasp renders at 50% nearly tripled the on-target rate to 7.2%, with a diagnostic type breakdown (11.3% on native frames vs. 3.1% on composites) that directly attributes the improvement to the restoration of refraction, shadow, and specular cues absent from composites. Data pipeline design determines whether VLA spatial learning occurs; data *fidelity*—specifically the presence of transparency-specific visual physics—determines how much of it occurs.

RQ2 (transparent-object perception). An RGB-only network recovers transparent-object geometry with 19.6° mean surface-normal error and 91% segmentation IoU within its training category. On categories never seen in training, segmentation degrades modestly ($\sim 77\%$ IoU) while normal estimation degrades in proportion to geometric dissimilarity from the training shapes (25.5° on a similar unseen object; 34.1° on a dissimilar one). Shape diversity in training data, rather than volume alone, governs geometric generalisation.

RQ3 (depth reconstruction). With perfect normal input, the Poisson solver statistically matches a strong inpainting baseline (0.136 m vs. 0.123 m RMSE; median difference 3.5 mm), validating the implementation; with predicted normals it reaches 0.416 m. The 0.28 m gap cleanly quantifies how perception error propagates into metric depth—the

dissertation’s sharpest single measurement—and identifies normal accuracy as the highest-leverage improvement in the geometry path. For the shallow object geometries evaluated, geometry-unaware inpainting is a competitive and vastly cheaper baseline, a boundary condition the source literature does not emphasise.

RQ4 (feasibility). The policy’s 27.6-bin mean spatial error ($\approx 11\%$ of image extent) supports coarse localisation but not reliable grasp targeting: 14.7% of predictions met a conservative grasp-proximity criterion (95% CI [10.4%, 20.3%]). The pipeline runs at 0.26 frames per second on an A100, bounded not by the 7-billion-parameter language model (35% of frame time) but by the CPU sparse depth solver (65%)—an inversion of the intuitive expectation with direct consequences for optimisation priorities.

[PLACEHOLDER — RQ5 conclusion on language-conditioned target selection, pending the multi-category semantic-reasoning experiment.]

6.2 Contributions to Knowledge and Practice

For the academic audience, the work contributes: a documented, diagnosed instance of shortcut memorisation specific to composited *action* supervision, together with the metric practices (per-coordinate error reporting; zero-loss-as-alarm) that expose it; a controlled error-propagation measurement linking normal-estimation error to metric depth error through a validated solver; and evidence on the shape-dependence of transparent-object geometric generalisation. For practitioners building sorting systems, the work delivers a costed map of the synthetic-only route: which data controls are non-negotiable, where a trivial inpainting baseline suffices and where it will not, and which pipeline stage to accelerate first for belt-rate operation.

6.3 Recommendations and Future Work

The findings motivate the following directions, ordered by expected return:

1. **Exact grasp measurement.** Replace the conservative proximity criterion with the mask-exact target-on-object test (requiring only that composited masks be saved at generation time). This corrects the headline feasibility figure at negligible cost. **[If completed before submission, this item moves into Chapter 4 and is removed here.]**
2. **Multi-category, instruction-conditioned training.** Extend ReSort-IT to all five ClearGrasp categories with per-category instructions, and evaluate instruction following on two-object scenes—testing whether the policy selects the *commanded* object. This directly addresses the semantic-flexibility motivation for using a VLA. **[In progress; if completed, this becomes Section 4.7 and is removed here.]**
3. **Data fidelity.** Replace alpha compositing with physically based rendering (or augment composites with synthetic refraction and contact shadows), and broaden the

shape distribution of training assets. Both the policy’s precision ceiling and the perception module’s OOD degradation trace to these gaps.

4. **Solver acceleration.** Port the sparse depth reconstruction to GPU (or amortise it into the perception network as a depth-completion head) to remove the 65% latency share and approach conveyor-rate throughput.
5. **Physical validation.** Close the loop on real hardware in staged steps: real RGB-D capture of transparent objects to measure the policy’s sim-to-real gap; open-loop reprojection accuracy in millimetres against a calibrated camera–robot transform; and only then live picking trials. Simulation of the ABB IRB 360 cell in Gazebo is a useful intermediate step if time permits.
6. **Spatially targeted supervision.** Incorporate pointing-style auxiliary data or spatial representations shown to sharpen VLA localisation [14, 15], attacking the 28-bin precision ceiling from the representation side as well as the data side.

6.4 Closing Remark

The value of this dissertation lies as much in what it measured as in what it built. Every major claim—that the data pipeline decides whether spatial learning occurs, that perception error costs 0.28 m of depth accuracy, that the solver and not the language model bounds throughput, that current precision is a factor of several short of grasp tolerance—is a number with a confidence interval or a significance test attached, obtained from a system whose failures were diagnosed as carefully as its successes. That evidentiary discipline, rather than any single component, is what the work most hopes to pass on.

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